

Spreading 44 Points in the Plane

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Abstract

We place 44 points in the plane so that the squared ratio of the largest to the smallest distance between them is at most 41.63370+. Erich Friedman’s table of best known configurations lists n up to 31. Deleting a point from our 45-point configuration gives 42.36644; this one is smaller. The ratio is computed exactly from the published coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 44 points in the plane with*

$$\left(\frac{\max d}{\min d}\right)^2 = 41.63370\dots$$

The exact value, a rational number, is given in Appendix A.

2 Context

Erich Friedman’s page *Minimizing the Ratio of Maximum to Minimum Distance* lists the best known configurations for $n \leq 31$. The value for $n = 44$ is not on the page. Deleting a point from our 45-point configuration gives 42.36644; this one is smaller.

3 Verification

The coordinates are decimals with 12 digits after the point, so they are exact rationals; all squared distances, and their largest-to-smallest ratio, are computed with integers. Run

```
python friedman-maxmin2d/verify.py friedman-maxmin2d/coords/n44.txt
```

in <https://github.com/Ge-limin/math>.

4 How it was found

Random starts of a constrained local optimizer (SLSQP), then basin hopping seeded from the neighbouring sizes (`friedman-maxmin2d/`).

A Exact value

```
numerator:
8326740155779162375330397
denominator:
199999999999674286549561
```

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

B Coordinates

44 points, x and y per line. File: friedman-maxmin2d/coords/n44.txt.

```
0.525385989252 11.480310453364
-3.087510144831 13.813584297956
-3.445493174489 11.635057186197
-0.198356873062 14.277883974387
-0.007380939123 10.560588611396
-4.098907765453 10.250074462288
-1.491392319755 7.956352505472
0.717053136290 13.875361285861
1.651190985750 9.666002909086
-2.021616898509 8.804209743505
-4.353919590845 11.217012396651
1.991944130748 10.606155712716
1.993913969728 12.374038699791
-0.005411100143 12.328471598472
-0.997708996907 12.452346067506
-1.006759135419 10.525329275502
-2.686554608268 9.551108560040
0.991997257174 10.595847947291
-2.217191664455 10.434113898062
-0.476534556665 9.677472037468
-2.473329563116 11.400754150105
-1.294797338956 13.407196064636

1.052867222129 8.864748471415
-2.993208881395 12.526931010357
-1.473370403341 11.409791781574
-2.174122574589 14.220675376866
0.053489025833 8.829489135521
-1.190654769825 14.401758443421
-3.665856812323 9.348705019721
-0.492005793445 7.991374941546
-3.802453882507 13.114402310232
-0.302499442192 13.283321595601
1.525332862826 11.490618218788
1.485013746226 13.234864212029
0.522843639631 9.712731373363
-3.189793381043 10.666715134940
-1.981176801671 12.271263000951
-1.552253954697 9.687215081527
-2.278265143720 13.226112998081
-3.000919102564 8.601806203187
-4.254738175600 12.222528486072
0.993967096153 12.363730934366
-0.473992207045 11.445051117469
-1.022029375943 8.839357843494
```